1 Where should the sensors go? Theory

The experiments of the previous section found that, among all layouts of six sensors, the best one is a regular ring whose radius depends on how directed the drone is. This section states the layout problem precisely, proves what can be proved about it, and separates those results from the numerical evidence for the rest.

1.1 The layout problem

Lengths are measured in sensor ranges ($\rho = 1$) and time in range-crossings at full speed ($v = 1$). The drone moves in the annulus $\Omega = \{x \in \mathbb{R}^2 : d < |x| < R\}$ by

$$dX_t = -\frac{X_t}{|X_t|}\,dt + \sqrt{2D}\,dW_t, \qquad D = 1/\mathrm{Pe}, \tag{1}$$

reflecting at the outer edge $|x| = R$ and arriving when it first reaches the response line $|x| = d$, at time τ. A layout is $s = (s_1, \dots, s_K) \in (\mathbb{R}^2)^K$, and detection is a Poisson process with rate

$$\kappa_s(x) = \mathrm{Da} \sum_{j=1}^{K} e^{-|x-s_j|^2/2}. \tag{2}$$

The chance of arriving undetected from x is $u_s(x) = \mathbb{E}_x\big[\exp(-\int_0^\tau \kappa_s(X_t)\,dt)\big]$, which by the Feynman–Kac formula solves

$$D\,\Delta u - \partial_r u - \kappa_s u = 0 \text{ in } \Omega, \qquad u = 1 \text{ on } |x| = d, \qquad \partial_r u = 0 \text{ on } |x| = R. \tag{3}$$

The drone chooses where to enter and the defender chooses the layout:

$$F(s) = \max_{|x|=R} u_s(x), \qquad F^* = \inf_s F(s). \tag{4}$$

A *regular ring* of radius a is $s_j = a\,e^{2\pi i j/K}$ (identifying $\mathbb{R}^2$ with $\mathbb{C}$, up to rotation); we write $F_{\mathrm{ring}}(a)$ and $a^* = \arg\min_a F_{\mathrm{ring}}(a)$. Throughout, $T_D(x) = \mathbb{E}_x\tau$ is the mean arrival time, which is finite and bounded on $\overline{\Omega}$.

1.2 Basic properties

Proposition 1 (Well-posedness)**.** *(a)* More detection never helps the drone: *if $\kappa_1 \le \kappa_2$ then $u_{\kappa_2} \le u_{\kappa_1}$, strictly on $\overline{\Omega} \setminus \{|x| = d\}$ if $\kappa_1 < \kappa_2$ on an open set.*

(b) Sensors belong inside the area: *replacing a sensor at $s_j = an$ with $a > R$, $|n| = 1$, by Rn strictly lowers u_s on the outer edge. Hence every minimizer lies in $(\overline{B}_R)^K$.*

(c) Continuity: $|F(s) - F(t)| \le \mathrm{Da}\,e^{-1/2}\,\sup_{\overline{\Omega}} T_D\,\sum_j |s_j - t_j|$.

(d) A minimizer of F exists, and F is invariant under rotations and reflections of the plane and under relabeling the sensors.

Proof. (a) follows path by path from the Feynman–Kac representation. (b) For $|x| \le R$, $|x - an|^2 - |x - Rn|^2 = (a - R)(a + R - 2x \cdot n) > 0$, so the moved sensor raises κ_s everywhere in Ω; apply (a) and compactness of the outer circle. (c) Coupling both layouts on the same path, $|e^{-A} - e^{-B}| \le |A - B| \le \tau\|\kappa_s - \kappa_t\|_\infty$, and the gradient of $e^{-|x-y|^2/2}$ in y has norm at most $e^{-1/2}$. (d) F is continuous on the compact set $(\overline{B}_R)^K$, which contains a minimizer by (b). □

1.3 Rings: where the drone gets through

For a ring, the detection rate at fixed distance r from the site is $\kappa(r,\theta) = \mathrm{Da}\, e^{-(r^2+a^2)/2} S_K(ra,\theta)$ with $S_K(z,\theta) = \sum_{j=0}^{K-1} e^{z\cos(\theta - 2\pi j/K)}$.

Lemma 2 (Coverage is thinnest midway between sensors)**.** *For $K \geq 2$ and $z > 0$, $S_K(z,\theta)$ is strictly decreasing in θ on $(0,\pi/K)$.*

Proof. For $0 < \theta < \pi/K$ put $c = \cos K\theta$, which decreases from 1 to -1. The distinct numbers $t_j = \cos(\theta - 2\pi j/K)$ are the roots of $T_K(t) - c$, where T_K is the Chebyshev polynomial with leading coefficient 2^{K-1}. Implicit differentiation gives $dt_j/dc = 1/T_K'(t_j)$, so with $f(t) = ze^{zt}$,

$$\frac{dS_K}{dc} = \sum_j \frac{f(t_j)}{T_K'(t_j)} = 2^{1-K} f[t_0,\ldots,t_{K-1}] = \frac{2^{1-K}}{(K-1)!} f^{(K-1)}(\xi) = \frac{2^{1-K} z^K e^{z\xi}}{(K-1)!} > 0,$$

using the divided-difference mean-value theorem. As c decreases with θ, so does S_K. □

Theorem 3 (The drone's best entry point)**.** *For every $D > 0$ and every ring radius $a > 0$, u on the outer edge is strictly increasing in θ on $(0,\pi/K)$. Hence $F_{\mathrm{ring}}(a) = u(R,\pi/K)$: the drone's best entry point is always midway between two neighboring sensors.*

Proof. Let $w = \partial_\theta u$ on the wedge $W = \{d < r < R,\ 0 < \theta < \pi/K\}$. The operator in (3) has coefficients independent of θ, so differentiating gives $D\Delta w - \partial_r w - \kappa w = (\partial_\theta \kappa)\, u$, which is negative in W by Lemma 2 and $u > 0$. By the ring's mirror symmetries $w = 0$ on $\theta = 0$ and $\theta = \pi/K$; $w = 0$ on $r = d$ since $u \equiv 1$ there; and $\partial_r w = 0$ on $r = R$. The strong minimum principle (the zeroth-order coefficient $-\kappa$ is nonpositive) and Hopf's lemma at the Neumann edge exclude a nonpositive minimum of w in W or on its outer arc, so $w > 0$ there. □

Proposition 4 (The best ring is a stationary point of the full problem)**.** *If $a > 0$ is a local minimizer of F_{ring}, then the ring of radius a is Clarke-stationary for F on $(\mathbb{R}^2)^K$: $0 \in \partial^C F(s(a))$.*

Proof. For $D > 0$, $u_s(R,\theta)$ is smooth in s, so by Danskin's theorem $C = \partial^C F(s(a)) = \mathrm{conv}\{\nabla_s u_s(R,\theta) : \theta \in A\}$, where A is the set of maximizing angles. With $e = s(a)/a$ the radial direction, local minimality along the ring gives $\max_{g\in C} g\cdot e \geq 0$ and $\min_{g\in C} g\cdot e \leq 0$, so some $h \in C$ has $h\cdot e = 0$. The dihedral group D_K acts on layouts by rotating or reflecting the plane and relabeling sensors; it fixes $s(a)$ and preserves C. Its fixed vectors are exactly the multiples of e: rotations force equal displacements in each sensor's own frame, and the reflection through a sensor removes the tangential part. The group average of h therefore lies in C and equals $(h\cdot e/K)\, e = 0$. □

1.4 The straight-line drone ($D \to 0$)

As $D \to 0$ the drone flies straight in along the ray at its entry angle, and $u_s(R,\theta) \to e^{-I_s(\theta)}$ with *exposure* $I_s(\theta) = \int_d^R \kappa_s(re^{i\theta})\, dr$. Minimizing F becomes maximizing $V(s) = \min_\theta I_s(\theta)$. Because V is proportional to Da, the best layout in this limit depends only on (K,d,R). One sensor at polar position (a,ϕ) contributes the closed form

$$h(a,\theta-\phi) = \sqrt{\tfrac{\pi}{2}}\, e^{-a^2\sin^2 t/2}\Big[\operatorname{erf}\tfrac{R-a\cos t}{\sqrt{2}} - \operatorname{erf}\tfrac{d-a\cos t}{\sqrt{2}}\Big], \qquad t = \theta - \phi. \tag{5}$$

Write $B_n(a) = \int_d^R e^{-(r^2+a^2)/2} I_n(ra)\, dr$ with I_n the modified Bessel function.

Proposition 5 (Exact ring exposure)**.** *For a ring of radius a, $I_{\text{ring}}(\theta) = K\mathrm{Da}\big[B_0(a)+2\sum_{q\geq 1} B_{qK}(a)\cos(qK\theta)\big]$, its minimum is at the midpoints, and*

$$M_K(a) := \min_\theta I_{\text{ring}} = K\mathrm{Da}\Big[B_0(a)+2\sum_{q\geq 1}(-1)^q B_{qK}(a)\Big], \qquad 0 \leq K\mathrm{Da}\, B_0(a) - M_K(a) \leq 2K\mathrm{Da}\, B_K(a).$$

Proof. Insert $e^{z\cos\varphi} = I_0(z) + 2\sum_{n\geq 1} I_n(z)\cos n\varphi$ and sum over the ring; only multiples of K survive. The minimum location is Lemma 2 integrated in r. The bound is the alternating-series estimate, since $I_n(z)$ decreases in n for $z > 0$. □

Proposition 6 (What the straight-line problem says)**.** *Let $V^* = \sup_s V(s)$ and $A_0 = \int_d^R e^{-r^2/2}dr$, $A_2 = \int_d^R (r^2-2)e^{-r^2/2}dr$.*

(a) Rings are asymptotically optimal: *$V^* \leq K\mathrm{Da}\max_a B_0(a)$, and the ring at $\arg\max_a B_0$ is within $2K\mathrm{Da}(R-d)\,(R^2/2)^K/K!$ of V^*.*

(b) Collapse to the center: *if $K \leq 2$, or if $R \leq 2$ and $A_2 \leq 0$, the unique optimum puts every sensor at the center.*

(c) No collapse: *if $K \geq 3$ and $A_2 > 0$, then $M_K(a) = K\mathrm{Da}A_0 + \frac{K\mathrm{Da}}{4}A_2a^2 + O(a^3)$, so a small ring beats the center.*

Proof sketch. (a) $V(s) \leq \frac{1}{2\pi}\int I_s = \mathrm{Da}\sum_j B_0(|s_j|)$; combine with Proposition 5 and $I_K(z) \leq (z/2)^K e^z/K!$. (b) For $K \leq 2$ some ray makes a nonacute angle with both sensors, and along it each sensor contributes at most what a central sensor would. For $R \leq 2$, $A_2 \leq 0$, the moment bound $\mathbb{E}_\mu r^{2m} \leq 2^m m!$ under $\mu \propto e^{-r^2/2}dr$ gives $B_0(a) < A_0$ for $a > 0$. (c) Expand $e^{-a^2/2}I_0(ra) = 1 + \frac{a^2}{4}(r^2-2) + O(a^4)$ and use $B_K = O(a^K)$. □

For the base geometry $d = 2$, $R = 6.5$ we have $A_2 > 0$, so collapse never occurs; the bound in (a) is only useful for large K. The straight-line best ring radius is $\arg\max_a M_K(a)$ (for $K = 6$: $a = 2.4528$, $M_6 = 1.30410$, $e^{-M_6} = 0.2714$), not $\arg\max_a B_0(a) = 3.060$, which maximizes *average* rather than worst-case coverage.

Proposition 7 (Local optimality of the hexagon; computer-assisted)**.** *For $K = 6$, $d = 2$, $R = 6.5$, the best ring radius is unique in $[2.452774, 2.452776]$, and the regular hexagon there is a strict local maximizer of V, modulo rotation.*

Status. Proved by an interval-arithmetic certificate of a second-order condition for max-min functions that accounts for movement of the worst entry angle (see the notes; **to be re-run from its code before submission**). Independently, all 800 random perturbations of size 3×10^{-3} and 10^{-2} that we tried lowered V.

Proposition 8 (Global certificate up to ε; branch and bound)**.** *For $K = 4$, $d = 2$, $R = 6.5$: no layout has $V(s) > M_4(a^*) + \varepsilon$, with $M_4(a^*) = 0.522161$ at $a^* = 1.822662$ and $\varepsilon = 2\times10^{-4}$ (0.04% of the optimum).*

Method. Branch and bound over the $2K-1$ layout coordinates (after fixing rotation, ordering the angles and putting the outermost sensor at angle 0; radii in $[0, R]$ by Proposition 1(b)). For a box of layouts and finitely many test angles θ, $\max V \leq \min_\theta \sum_j \sup_{\text{box}} h$; the supremum over each sensor's angular range is attained at the closest angle because h decreases in $|t|$, and over its radial range by tabulated values of (5) plus the bound $|\partial_a^2 h| \leq 5.5$. Boxes whose bound is at most the

target are discarded; the rest are halved. Near the ring this first-order bound loses accuracy in proportion to the box width, so small boxes also use a second-order bound: a Taylor expansion of each sensor's contribution with a rigorous remainder (Hessian and gradient bounds that shrink with the sensor's distance from each test ray), combined over test angles by linear-programming dual weights. The final bound is evaluated from those weights directly, so it is valid whatever the solver's accuracy. The $K = 4$ certificate took 3.2×10^6 boxes and 56 minutes on one core (at $\varepsilon = 10^{-3}$: 10^5 boxes, 90 seconds). For $K = 6$ the same method did not finish: after 90 minutes, even with the weak target $\varepsilon = 0.1$, about a quarter of the search space remained open. The bottleneck is the five free angles. A box that allows each sensor a wide range of angles is bounded as if every sensor sat at the best angle for each ray, which behaves like a ring smeared into a continuous band (exposure up to $K\mathrm{Da}\max_a B_0 = 1.667$, well above the hexagon's 1.304); only boxes that are narrow in all five angles can be discarded. Certifying $K = 6$ will need a bound that keeps sensors discrete across rays, or a symmetry reduction, and is left for future work.

1.5 How the best radius depends on drone randomness

Proposition 9 (Both ends of the randomness scale)**.** *(a)* Nearly straight drones (formal expansion): $F_{\mathrm{ring}}(a) = e^{-M_K(a)}\big[1 + D\,H_K(a) + O(D^2)\big]$ *with*

$$H_K(a) = \kappa(d, \tfrac{\pi}{K}) + \int_d^R \Big[\kappa^2 - \frac{\kappa}{r} - \frac{J_{\theta\theta}}{r^2}\Big]_{\theta=\pi/K} dr, \qquad J(r,\theta) = \int_d^r \kappa(q,\theta)\, dq,$$

the term $\kappa(d, \pi/K)$ coming from the boundary layer at the reflecting edge. Hence $a^*(D) = a^*(0) + cD + O(D^2)$ with $c = H_K'(a^*(0))/M_K''(a^*(0))$; for the base case $c \approx 3.317$.

(b) Strongly wandering drones: $u_s = 1 - w_s/D + O(D^{-2})$ *where $-\Delta w_s = \kappa_s$, $w_s = 0$ at $r = d$, $\partial_r w_s = 0$ at $r = R$; the best ring radius tends to the maximizer of $\min_\theta w_{s(a)}(R,\theta)$, which for the base case is $a_\infty \approx 4.96$.*

Table 1: Best ring radius for six sensors ($d = 2$, $R = 6.5$, $\mathrm{Da} = 1$): finite-difference solutions of (3) against the two asymptotic predictions.

D	0	0.005	0.01	0.02	0.05	0.5	2	8	32	∞
PDE	2.453	2.471	2.487	2.517	2.599	3.25	4.03	4.67	4.89	
$a^*(0) + cD$		2.469	2.486	2.519	2.619					
Large-D limit										4.96

The best radius grows steadily with randomness, from the straight-line value 2.453 toward 4.96: a defender facing wandering drones should push the ring outward, where it covers the longer, more tortuous approach paths. Monotonicity for all D is supported by Table 1 but not proved.

1.6 A floor no layout can beat

Proposition 10 (Rotational-average lower bound)**.** *For every layout, $F(s) \geq u_{\bar\kappa_s}(R)$, where $\bar\kappa_s(r) = \mathrm{Da}\sum_j e^{-(r^2+|s_j|^2)/2} I_0(r|s_j|)$ is κ_s averaged over rotations. Consequently $F^* \geq \min_{a_1,\ldots,a_K \geq 0} u_{\bar\kappa(a)}(R)$, a problem with radial symmetry. If $R \leq \sqrt{2}$ the bound is attained by putting every sensor at the center.*

Proof. By Jensen's inequality path by path, $u_{\bar\kappa_s}(x) \le \frac{1}{2\pi}\int_0^{2\pi} u_{Q_\phi s}(x)\,d\phi$, and by rotational invariance of (1) the right side at $|x| = R$ is the average of u_s over entry angles, which is at most $F(s)$. For $R \le \sqrt{2}$, $e^{-a^2/2} I_0(ra) \le e^{-a^2/2 + r^2a^2/4} \le 1$, so $\bar\kappa_s \le K\mathrm{Da} e^{-r^2/2}$. □

Table 2: Six sensors, $d = 2$, $R = 6.5$, Da $= 1$: the best ring against the floor of Proposition 10 (radial problem solved numerically; the minimizing radii were all equal).

Drone	Pe	Best ring F	Floor	Gap
Wandering	0.5	0.4440	0.4099	0.034
Mixed	2	0.3101	0.2683	0.042
Directed	20	0.2710	0.1986	0.072

The floor is tightest for wandering drones, which effectively average the detection field over angle. Averaging only over the ring's own K-fold rotations does not tighten it: the averaged field of six suitably placed sensors is that of a 36-sensor ring of one-sixth strength, which is within 10^{-9} of the fully averaged field.

1.7 Is the ring the best layout? Evidence and conjecture

Conjecture 11. *For $d = 2$, $R = 6.5$, Da $= 1$ and $K = 6$, every minimizer of F is a regular ring, for every $D > 0$.*

Evidence, beyond Propositions 4, 7 and 8:

- Gradient search (exact adjoint gradients of a soft maximum of the edge values, L-BFGS-B) from 36 starts per drone type, 30 of them random: all 108 runs ended at a regular hexagon (spacing 60°, radii within 0.03), within 2×10^{-4} of the best ring, below the grid error.
- Straight-line limit: 45 epigraph searches (SLSQP), 40 from random layouts, all ended at the hexagon of radius 2.453; for $K = 4$, likewise at the square of radius 1.823.
- Structured families (two staggered rings, zigzag, five plus one, four plus two) either collapse to the ring or are worse by 0.08–0.20; a seeded evolutionary search never improved on its ring seed.
- Pattern changes: for straight-line drones the single ring beats ring-plus-center and every two-ring split for $K = 3, \dots, 10$, unlike the small-trap eigenvalue problem, where a ring plus a center trap wins from seven traps [1].

Conjecture 11 cannot hold for every geometry: by Propositions 6(b) and 10, small areas ($R \le \sqrt{2}$, or $R \le 2$ with $A_2 \le 0$ for straight-line drones) are best defended by stacking every sensor at the center. The ring is the answer when the area is large compared with a sensor's range.

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